

Chris KAMAL YOUSSEF

Freelance: Computer Vision & AI Engineer | Intelligent Systems

Location: Paris, France | Open to worldwide

[in LinkedIn](#) | ©0658610499 | [✉ chris11499@outlook.com](mailto:chris11499@outlook.com) | Nationality:  Italian

Skills

- Programming: Python | C++ | C | Typescript | Object-Oriented Programming | Design Patterns | Git | Linux
- Image processing and Computer Vision | Neural Networks | XR | Signal processing | Robotics | Embedded Systems
- ROS 2 | Gazebo | RViz | Arduino | Unity || OpenCV | Pytorch | Tensorflow | Roboflow | Ultralytics | Pandas
- French (bilingual) | English (professional)

Professional Experience

- | | | | |
|--|--|------------------------------------|--------------------------|
| Computer Vision : freelance | <u>Football federation</u> | <i>Remote</i> | 11/2025 - Current |
| • Computer Vision and Machine Learning to analyze players performance during football game | | | |
| Computer Vision : freelance | <u>Nicomatic France</u> | <i>Paris, France</i> | 07/2025 - Current |
| SkyAI Project, autonomous drone without GPS, run real time on Raspberry Pi 4 | | | |
| • Visual Odometry pipeline : keypoint detection, feature matching, affine transform estimation, and projection on map tiles. | | | |
| • Tracking objects on ground detecting direction and speed while the drone is moving irregularly . | | | |
| Crypto programmer : freelance | <u>Candide Atelier</u> | <i>Remote</i> | 04/2025 - 07/2025 |
| • Developing libraries that help developers create secure digital wallets and cover users' gas fees. Typescript | | | |
| Robotics, AI and Computer Vision : contract | <u>Shark Robotics</u> | <i>La Rochelle, France</i> | 08/2024 - 02/2025 |
| • Lead autonomy development for Firefighting AMR (Visual Odometry, LiDAR, Ultrasound, CV) | | | |
| • Neural network training and Computer Vision for different perception tasks | | | |
| • Managed sensor suite (cameras, LiDAR, ultrasound) across robot fleet | | | |
| • Coding in Python, ROS 2 | | | |
| Python Developer : contract | <u>AIRBUS</u> | <i>Toulouse, France</i> | 02/2024 - 08/2024 |
| • Refactored legacy Python codebase (tens of large scripts) into a modular, maintainable library used across teams | | | |
| • Applied Design Patterns (Builder, Strategy) to improve readability and scalability | | | |
| Computer Vision engineer | <u>ALTEN Labs</u> | <i>Toulouse, France</i> | 10/2023 - 02/2024 |
| Contribute to the Smart Factory 4.0 research project | | | |
| • Optimizing Computer Vision algorithms and Neural Networks for object detection and segmentation | | | |
| Information processing : Internship | <u>INERIS</u> | <i>Verneuil-en-Halatte, France</i> | 02/2023 - 08/2023 |
| DESIHR Research Project: Autonomous Drones for Real-Time Pollution Monitoring | | | |
| • Developed image processing and neural network models (YOLOv5, RAFT) for accurate smoke and flame data extraction used to calculate pollution. Achieving 92% precision in detection . 0.067 mean square error in optical flow analysis . | | | |
| • Programmed STM32 boards , designing digital filters that reduced rise time from 71ms to 21ms with only 1% overshoot and no noise . | | | |
| • Organization and participation in 3-day tests in Normandy to validate drone deployment under realistic conditions. | | | |
| Signal processing : Internship | <u>Ecole Polytechnique Montréal</u> | <i>Quebec, Canada</i> | 05/2022 - 08/2022 |
| • Automate data flow using MATLAB and Python , from data collection, signal filtering to processing and storage. | | | |

Education

- | | | | |
|---|---|-------------------------|--------------------------|
| Master's degree: Automatic, Robotics: Intelligent Systems Engineering (ISI) | <u>Sorbonne University</u> | <i>Paris, France</i> | 09/2021 - 09/2023 |
| Master 2 option: "Information processing" :
Image, Sound and Signal processing. AI and Deep Neural Networks. Biometrics. Virtual Reality. Bio-inspired Vision. Robotics ROS 2. | | | |
| Bachelor's degree: Electronics, Electrical Energy, Automatic (EEA) | <u>Grenoble-Alpes University</u> | <i>Grenoble, France</i> | 09/2018 - 07/2021 |

Competitions

- **SensAI and Meta XR Hackathon: First prize** In 2025. "AI with Camera access". Was responsible for the Computer Vision part. Built an XR car-repair assistant on Meta Quest 3 in 48h (Unity + Roboflow), recognized by Meta, Unity & partners.
- **French Robotics Cup**: In 2020 on FPGA via VHDL | In 2021 on STM32 using C
Coding the: Motor and speed control, color sensor, Obstacle detector, the path to be realized by the robot.
- **Paris Autonomous Car Cup**: In 2022 ROS 2 | Navigate autonomously along a trajectory, avoiding different obstacles.
In simulation on **Gazebo/Rviz** and on a real **Turtlebot3 Burger**.

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Humanitarian mission

- **South Sudan:** Living with them and identifying their needs for future missionaries **(08/2017 - 10/2017)**

Coursera Certificates and Projects

- **Self-Driving Cars:** 4 courses. Using CARLA, covering hardware, software, state estimation, visual perception, and motion planning.
- **Semantic Segmentation with Amazon Sagemaker:** Project
- **Hyperparameter Tuning with Neural Network Intelligence:** Project

Interests

Travel | Chess | Bodybuilding | Field Hockey | Arche Community (Welcoming mentally handicapped people)

Projects

- **Image Processing:**
 - **Gym Workout Coach:** Developed an AI coach for real-time workout monitoring. Using **MediaPipe**. Counting reps, and assessing efficiency based on range of motion and other muscles movement.
 - **AI Cardiology:** Implemented a Kalman filter to track myocardial movement in ultrasound image sequences, enhancing diagnostic accuracy. Using **OpenCV**
- **Machine Learning Projects:**
 - **Fader Networks:** Implemented code of "Fader Networks: Manipulating Images by Sliding Attributes" using Auto-Encoders and GANs to adjust image attributes dynamically. Used **Tensorflow**.
 - **Conditional GAN Image Generator:** Developed a Conditional GAN model to generate realistic images based on varying input conditions. Used **Tensorflow**.
 - **Celebrity Photo Classifier:** Programmed multiple CNN models to classify celebrity photos with high accuracy, training on a large dataset. Used **Pytorch**.
- **C++ Projects:**
 - **Transportation app:** Back end of app that finds the shortest route between two metro stations using the DIJKSTRA algorithm and Paris metro data.